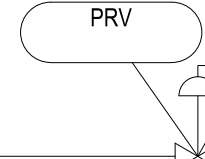
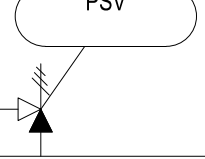
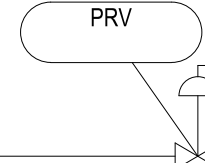
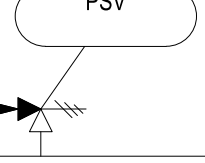
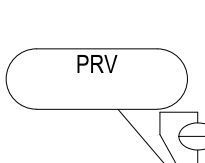
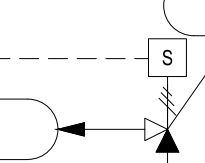
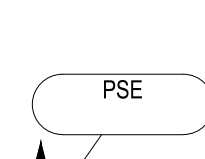
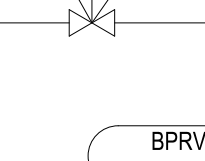
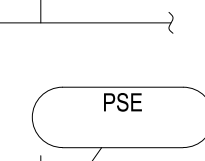


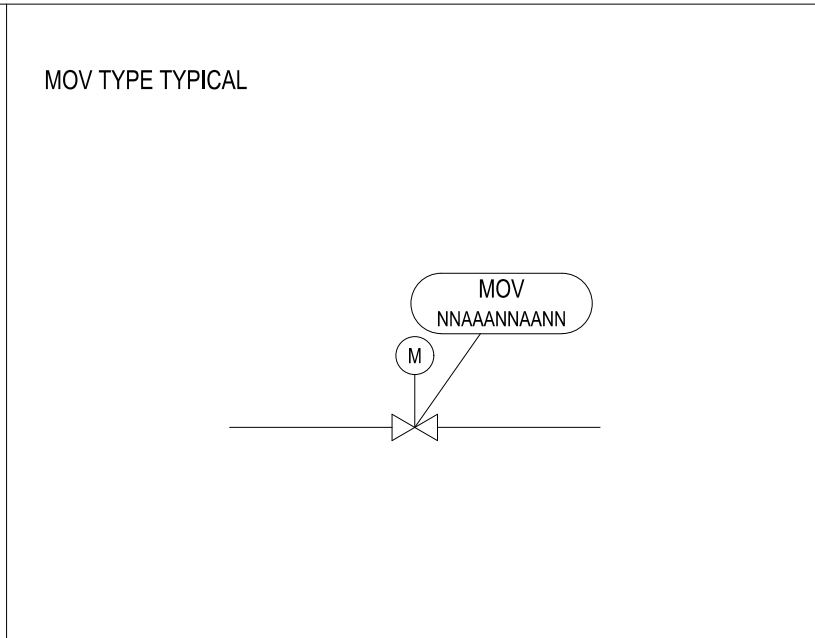
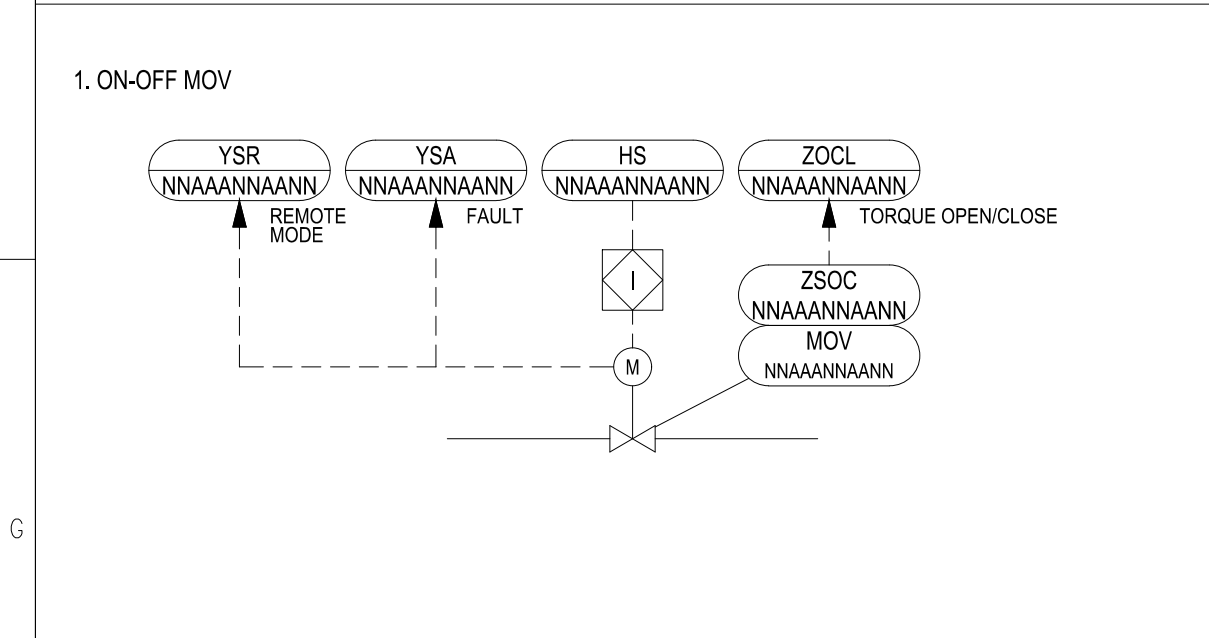
[illegible]



|                     |                                    |                    |             |                |          |                       |                   |                |        |                          |            |                      |                             |                  |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
|---------------------|------------------------------------|--------------------|-------------|----------------|----------|-----------------------|-------------------|----------------|--------|--------------------------|------------|----------------------|-----------------------------|------------------|-------------------------|------------------------------|------------|---------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
| 8                   |                                    | 7                  |             |                | 6        |                       |                   | 5              |        |                          | 4          |                      |                             | 3                |                         |                              | 2          |               |              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |                                                                                                                                             |  |  |  |
| FIRST LETTER        |                                    | SUCCEEDING LETTERS |             |                |          |                       |                   |                |        |                          |            |                      |                             |                  |                         |                              |            |               |              | SELF-ACTUATED DEVICES-PRESSURE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  | GENERAL NOTES :<br><br>1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE. |  |  |  |
|                     |                                    | SENSING DEVICE     |             | DISPLAY DEVICE |          |                       |                   | CONTROL DEVICE |        |                          |            | MISCELLANEOUS DEVICE |                             |                  |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| SYMBOL              | MEASURED OR INITIATING VARIABLE    | PRIMARY ELEMENT    | TRANSMITTER | INDICATOR      | RECORDER | INTEGRATING INDICATOR | ARARM ANNUNCIATOR |                |        | CONTROL STATION MAN/AUTO | CONTROLLER | CONTROL VALVE        | OTHER FINAL CONTROL ELEMENT | SWITCH           | LOCAL OBSERVATION GLASS | SOLENOID RELAY OR POSITIONER | TEST POINT | USER'S CHOICE | UNCLASSIFIED | NOTES :<br><br> PRESSURE REDUCING REGULATOR SELF CONTAINED<br> PRESSURE RELIEF OR SAFETY VALVE ANGLE PATTERN SPRING OR WEIGHT LOADED OR WITH INTEGRAL PILOT VALVE OR WITH INTEGRAL PILOT VALVE<br> PRESSURE REDUCING REGULATOR WITH EXTERNAL PRESSURE TAP<br> VACUUM RELIEF VALVE ANGLE PATTERN SPRING OR WEIGHT LOADED OR WITH INTEGRAL PILOT VALVE<br> DIFFERENTIAL PRESSURE REDUCING REGULATOR WITH INTERNAL AND EXTERNAL PRESSURE TAPS WITH INTERNAL AND EXTERNAL PRESSURE TAPS<br> PRESSURE RELIEF OR SAFETY VALVE BY INTEGRAL SOLENOID<br> BACKPRESSURE REGULATOR SELF-CONTAINED<br> RUPTURE DISC OR SAFETY VALVE FOR PRESSURE RELIEF<br> BACKPRESSURE REGULATOR WITH EXTERNAL PRESSURE TAP<br> RUPTURE DISC OR SAFETY HEAD FOR VACUUM RELIEF |  |  |  |                                                                                                                                             |  |  |  |
| ( )                 | TYPICAL SYMBOL                     | ( )E               | ( )T        | ( )I           | ( )R     | ( )QI                 | ( )AL             | ( )AH          | ( )AHL | ( )K                     | ( )C       | ( )CV                | ( )Z                        | ( )S             | ( )G                    | ( )Y                         | ( )P       | ( )N          | ( )X         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| A                   | ANALYSIS                           | A E                | A T         | A I            | A R      |                       | AAL               | AAH            | AAHL   | A K                      | A C        |                      | A Z                         | A S              |                         | A Y                          | A P        | A N           | A X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| B                   | BURNER FLAME                       | B E                | B T         | B I            | B R      |                       | BAL               |                |        |                          | B C        |                      |                             | B S              |                         | B Y                          | B P        | B N           | B X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| C                   | CONDUCTIVITY                       | C E                | C T         | C I            | C R      |                       | CAL               | CAH            | CAHL   | C K                      |            |                      | C Z                         | C S              |                         | C Y                          | C P        | C N           | C X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| D                   | DENSITY (MASS) OR SPECIFIC GRAVITY | D E                | D T         | D I            | D R      |                       | DAL               | DAH            | DAHL   | D K                      | D C        |                      | D Z                         | D S              |                         | D Y                          | D P        | D N           | D X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| E                   | VOLTAGE (EMF)                      | E E                | E T         | E I            | E R      |                       | EAL               | EAH            | EAHL   | E K                      | E C        |                      | E Z                         | E S              |                         | E Y                          |            | E N           | E X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| F                   | FLOW                               | F E                | F T         | F I            | F R      | FQI                   | FAL               | FAH            | FAHL   | F K                      | F C        | FCV                  | F Z                         | F S              | F G                     | F Y                          | F P        | F N           | F X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| G                   | GAUGING (DIMENSIONAL)              | G E                | G T         | G I            | G R      |                       | GAL               | GAH            | GAHL   | G K                      | G C        |                      |                             | G S              |                         | G Y                          |            | G N           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| H                   | HAND (MANUAL INITIATED)            | H                  |             |                |          |                       |                   |                |        |                          | H C        |                      | H Z                         | H S              |                         | H Y                          |            | H N           |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| I                   | CURRENT                            | I E                | I T         | I I            | I R      | IQI                   | IAL               | IAH            | IAHL   | I K                      |            |                      | I Z                         | I S              |                         | I Y                          |            | I N           | I X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| J                   | POWER                              | J E                | J T         | J I            | J R      | JQI                   | JAL               | JAH            | JAHL   | J K                      |            |                      | J Z                         | J S              |                         | J Y                          |            | J N           | J X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| K                   | TIME                               | K E                | K T         | K I            | K R      | KQI                   | KAL               | KAH            | KAHL   |                          | K C        |                      | K Z                         | K S              |                         | K Y                          |            | K N           | K X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| L                   | LEVEL                              | L E                | L T         | L I            | L R      |                       | LAL               | LAH            | LAHL   | L K                      | L C        | LCV                  | L Z                         | LS               | L G                     | L Y                          | LP         | LN            | L X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| M                   | MOISTURE OR HUMIDITY               | M E                | M T         | M I            | M R      |                       | MAL               | MAH            | MAHL   | M K                      | M C        |                      | M Z                         | MS               |                         | M Y                          | MP         | M N           | M X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| O                   | USER'S CHOICE                      |                    |             |                |          |                       |                   |                |        |                          |            |                      |                             |                  |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| P                   | PRESSURE OR VACUUM                 | P E                | P T         | P I            | P R      |                       | PAL               | PAH            | PAHL   | P K                      | P C        | PCV                  | P Z                         | P S              |                         | P Y                          | P P        | P N           | P X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| PD                  | PRESSURE DIFFERENTIAL              | PDE                | PDT         | PDI            | PDR      |                       | PDAL              | PDH            | PDHL   | PDK                      | PDC        |                      | PDZ                         | PDS              |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| Q                   | QUANTITY                           | Q E                | Q T         | Q I            | Q R      | QQI                   | QAL               | QAH            | Q AHL  | Q K                      | Q C        |                      | Q Z                         | Q S              |                         | Q Y                          |            | Q N           | Q X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| S                   | SPEED OR FREQUENCY                 | S E                | S T         | S I            | S R      | SQI                   | SAL               | SAH            | SAHL   | S K                      | S C        |                      | S Z                         | SS               |                         | S Y                          |            | S N           | S X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| T                   | TEMPERATURE                        | T E                | T T         | T I            | T R      |                       | TAL               | TAH            | TAHL   | T K                      | T C        | TCV                  | T Z                         | T S              |                         | T Y                          | TP         | T N           | T X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| U                   | MULTIVARIABLE                      |                    |             | U I            | U R      |                       |                   |                |        |                          |            |                      | U Z                         | U S              |                         | U Y                          |            | U N           | U X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| V                   | VIBRATION                          |                    | V T         | V I            | V R      |                       | VAL               | VAH            | VAHL   | V K                      | V C        |                      | V Z                         | VS               |                         | V Y                          | VP         | V N           | V X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| W                   | WEIGHT OR FORCE                    | W E                | W T         | W I            | W R      | WQI                   | WAL               | WAH            | WAHL   | W K                      | W C        |                      | W Z                         | WS               |                         | W Y                          |            | W N           | W X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| X                   | UNCLASSIFIED                       |                    |             |                |          |                       |                   |                |        |                          |            |                      |                             |                  |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| Y                   | EVENT OR STATE                     | Y E                | Y T         | Y I            | Y R      |                       | YAL               | YAH            |        |                          |            |                      | Y Z                         | Y S              |                         | Y Y                          |            | Y N           | Y X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| Z                   | POSITION                           | Z E                | Z T         | Z I            | Z R      |                       | ZAL               | ZAH            | Z AHL  |                          |            |                      | Z Z                         | Z S              |                         | Z Y                          |            | Z N           | Z X          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
| GENERAL INSTRUMENTS |                                    | SENSOR-FLOW        |             |                |          | SENSOR-FLOW (CONT.)   |                   |                |        | SENSORS-PRESSURE         |            |                      |                             | VALVES ACTUATORS |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |
|                     |                                    |                    |             |                |          |                       |                   |                |        |                          |            |                      |                             |                  |                         |                              |            |               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  |                                                                                                                                             |  |  |  |



|   |                                                            |   |   |
|---|------------------------------------------------------------|---|---|
|   | 8                                                          | 7 | 6 |
| H | VALVE BODY WITH ACTUATOR-TYPICAL FOR MOTOR OPERATED VALVES |   |   |

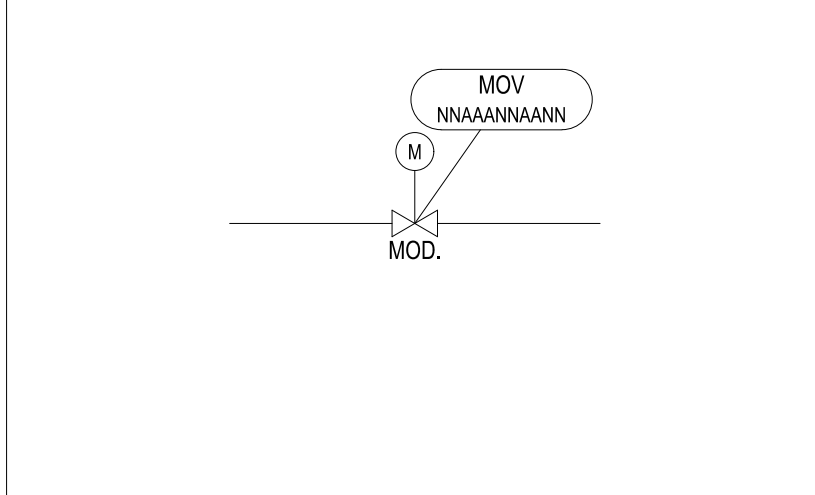
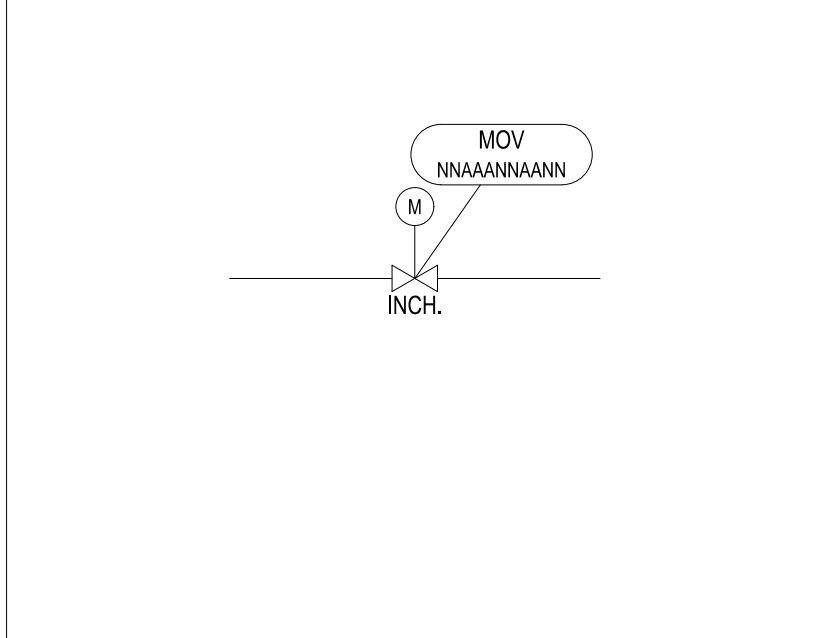


### 3. MODULATING

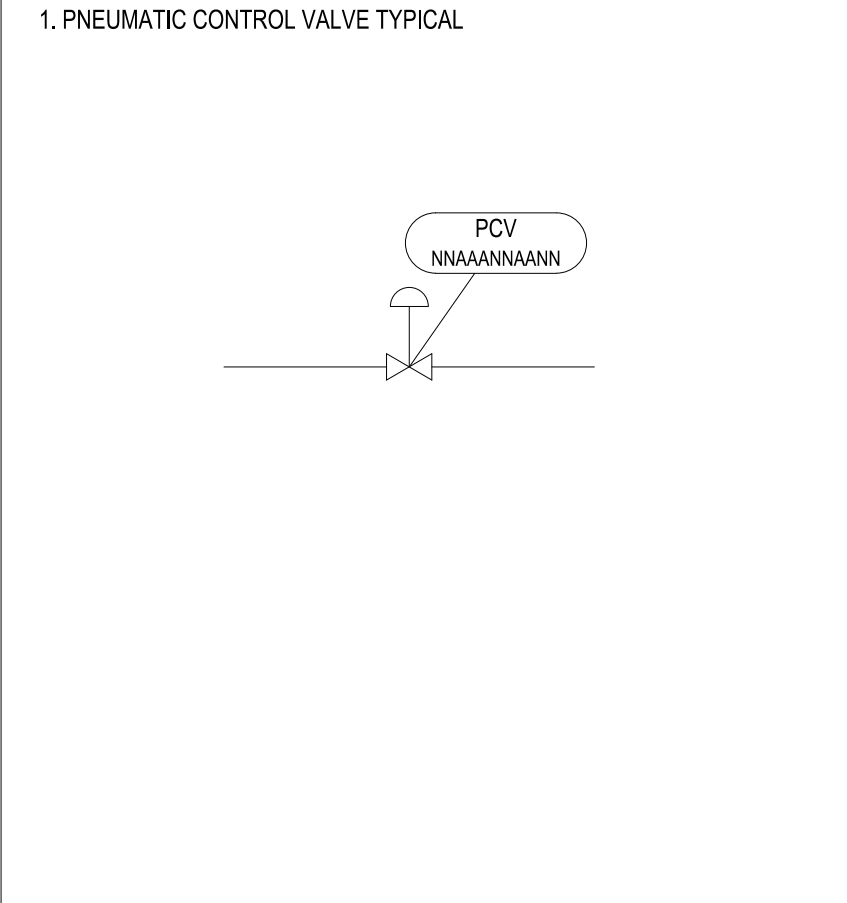
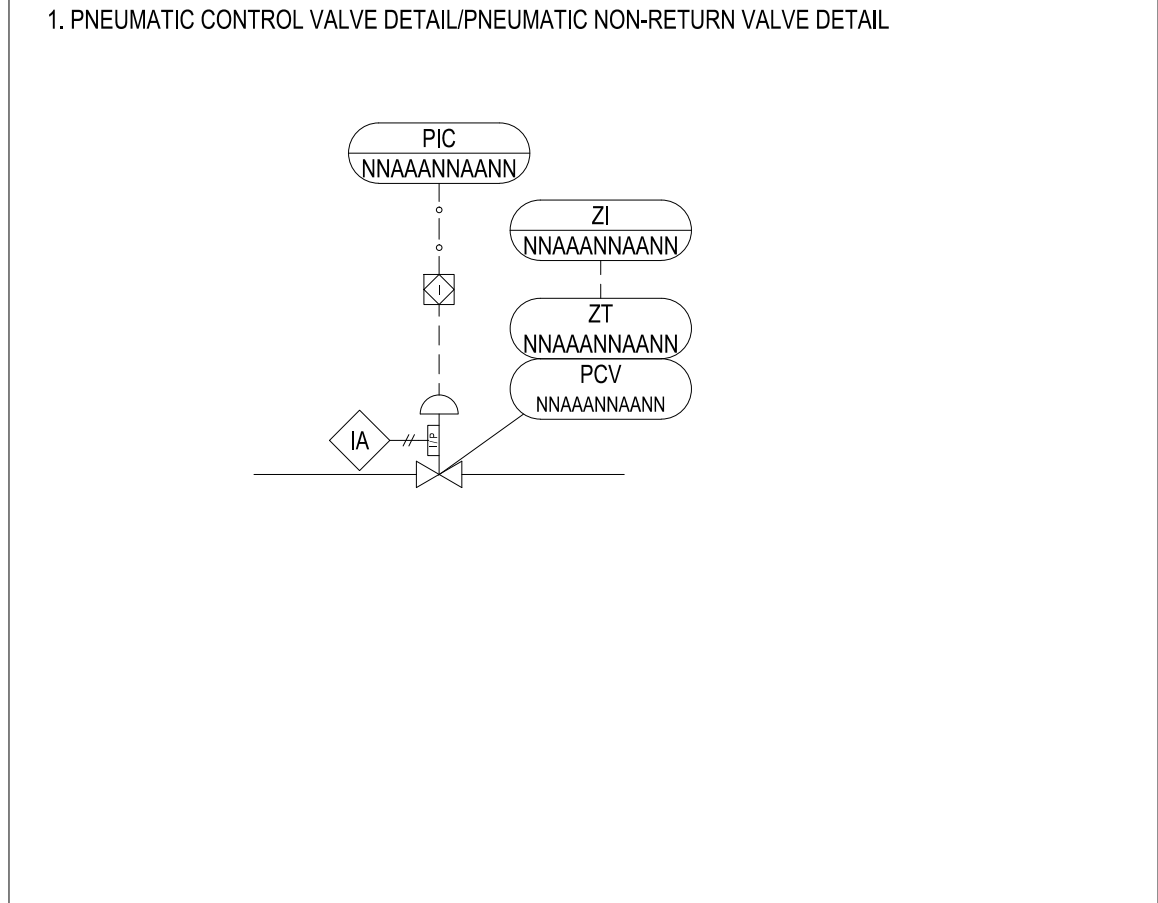
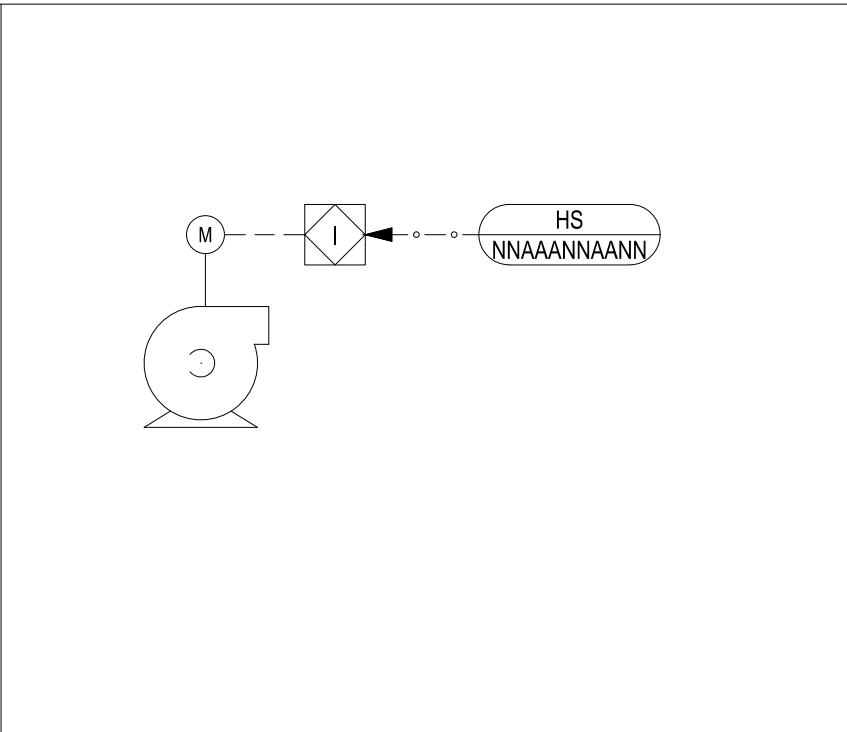
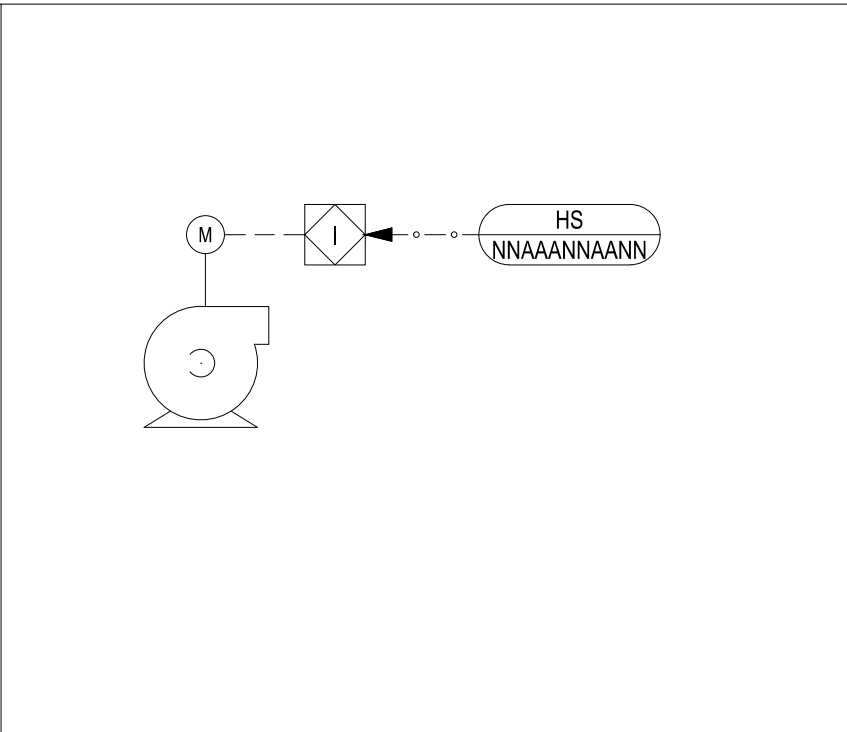
The diagram illustrates the Modulating stage of a 4-stage pipeline. It consists of four registers, each containing the value NNAAANNAANN:

- YSR** (Remote Mode): Contains NNAAANNAANN. A dashed line connects it to the YSA register.
- YSA** (Fault): Contains NNAAANNAANN. A dashed line connects it to the YSR register.
- ZIC** (I): Contains NNAAANNAANN. A solid line connects it to a multiplexer (M) labeled MOD.
- ZI** (ZT, MOV): Contains NNAAANNAANN. A solid line connects it to the multiplexer (M) labeled MOD.

The multiplexer (M) is labeled MOD. and its output is connected to the ZI register.

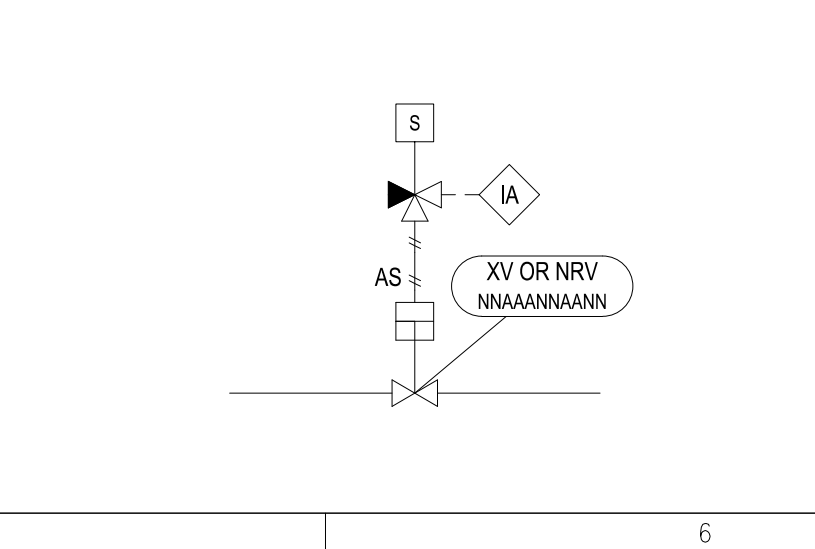
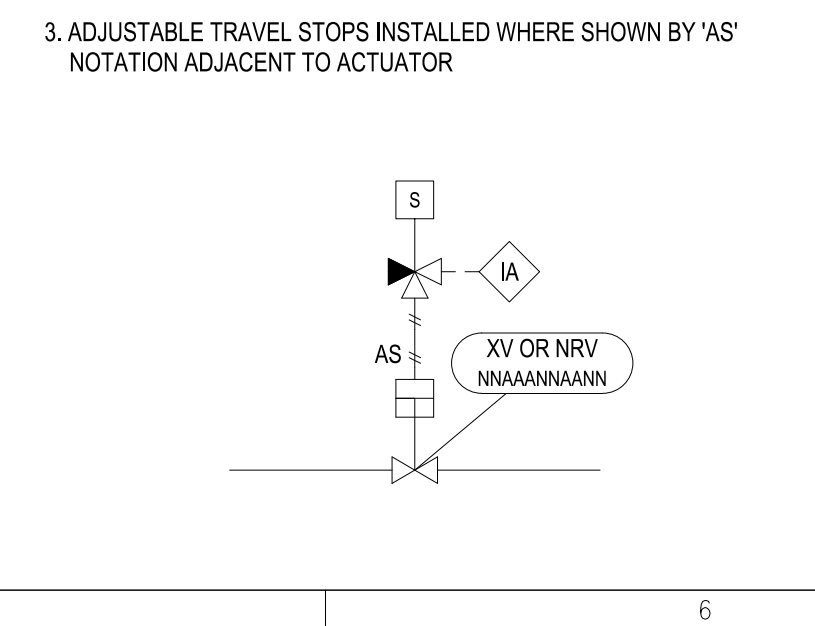
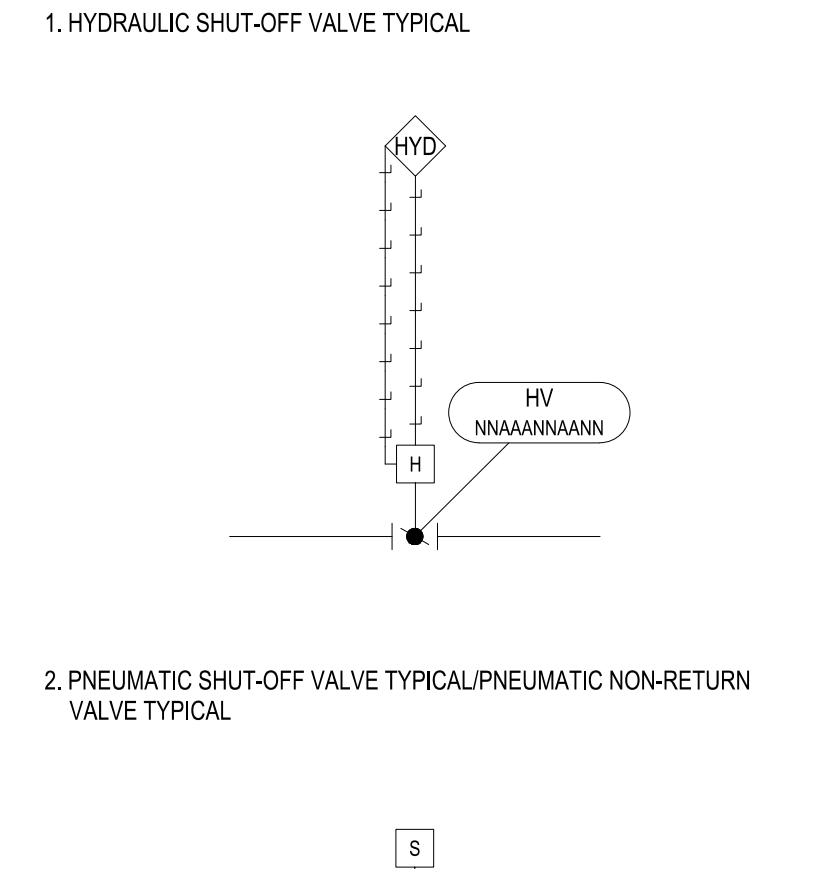
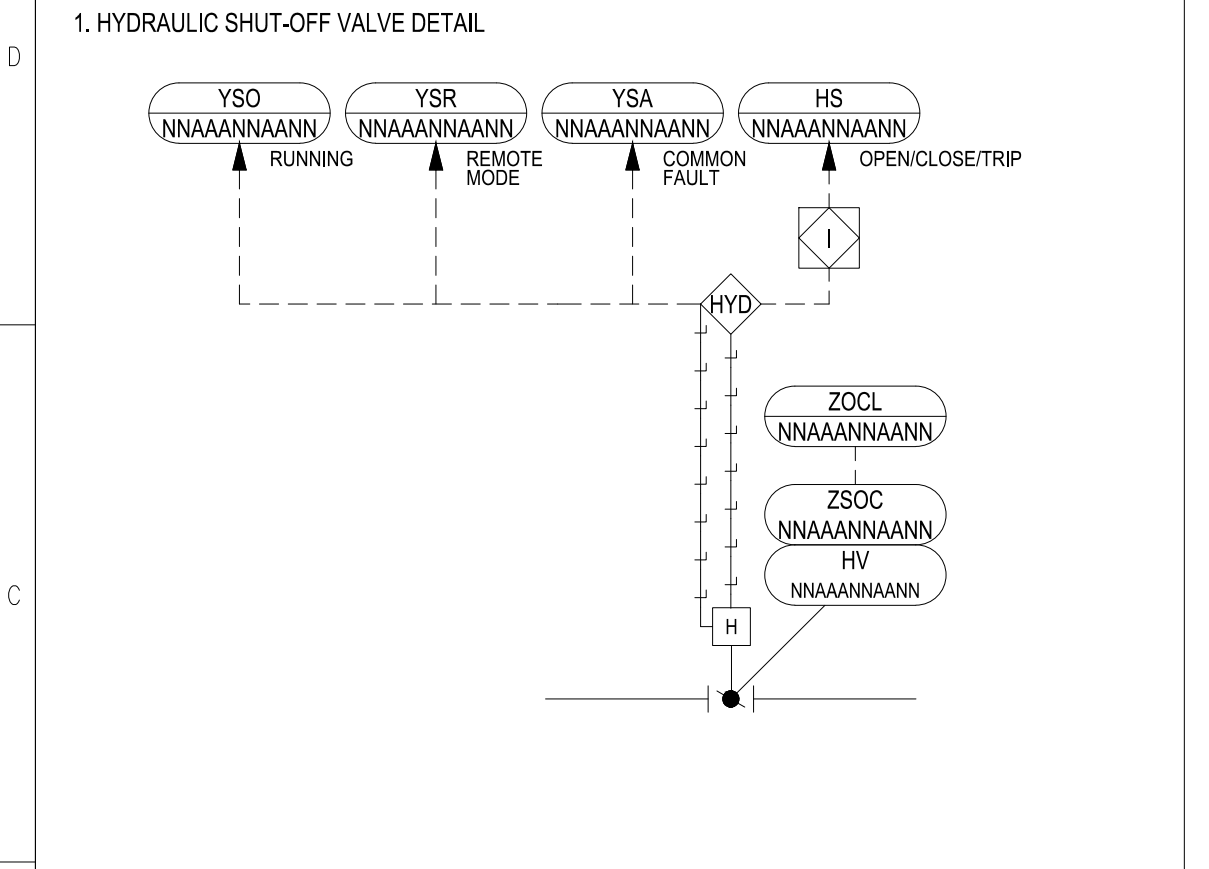


MOTOR WITHOUT DUTY/STANDBY SELECTION (TYP.1)

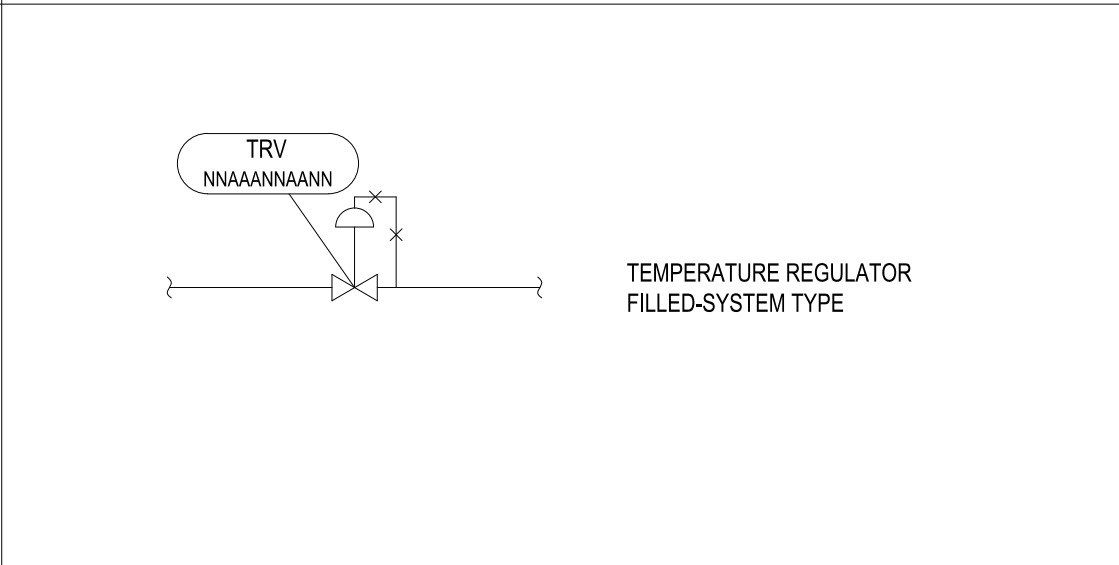


|  |  |
|--|--|
|  |  |
|  |  |

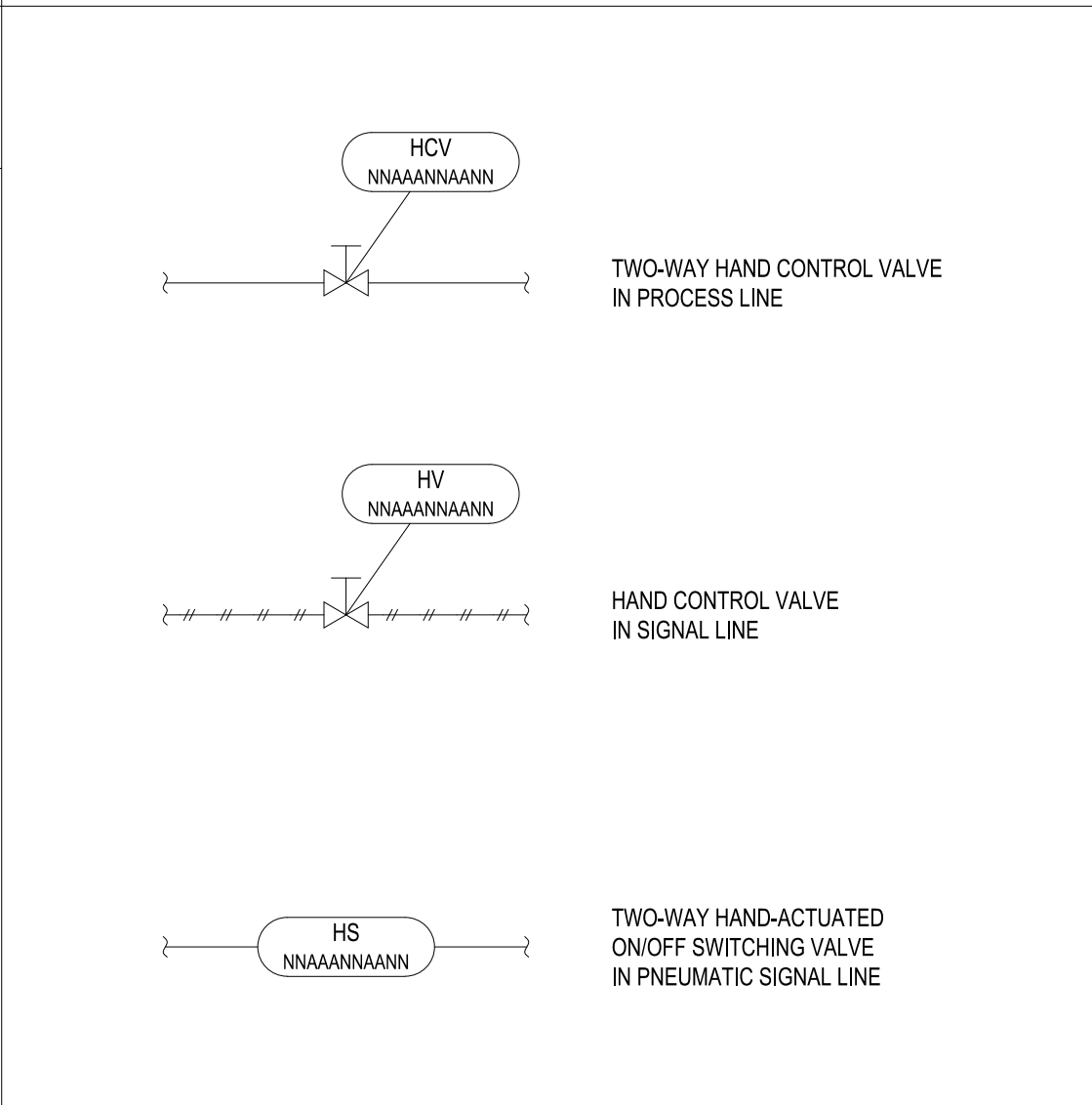
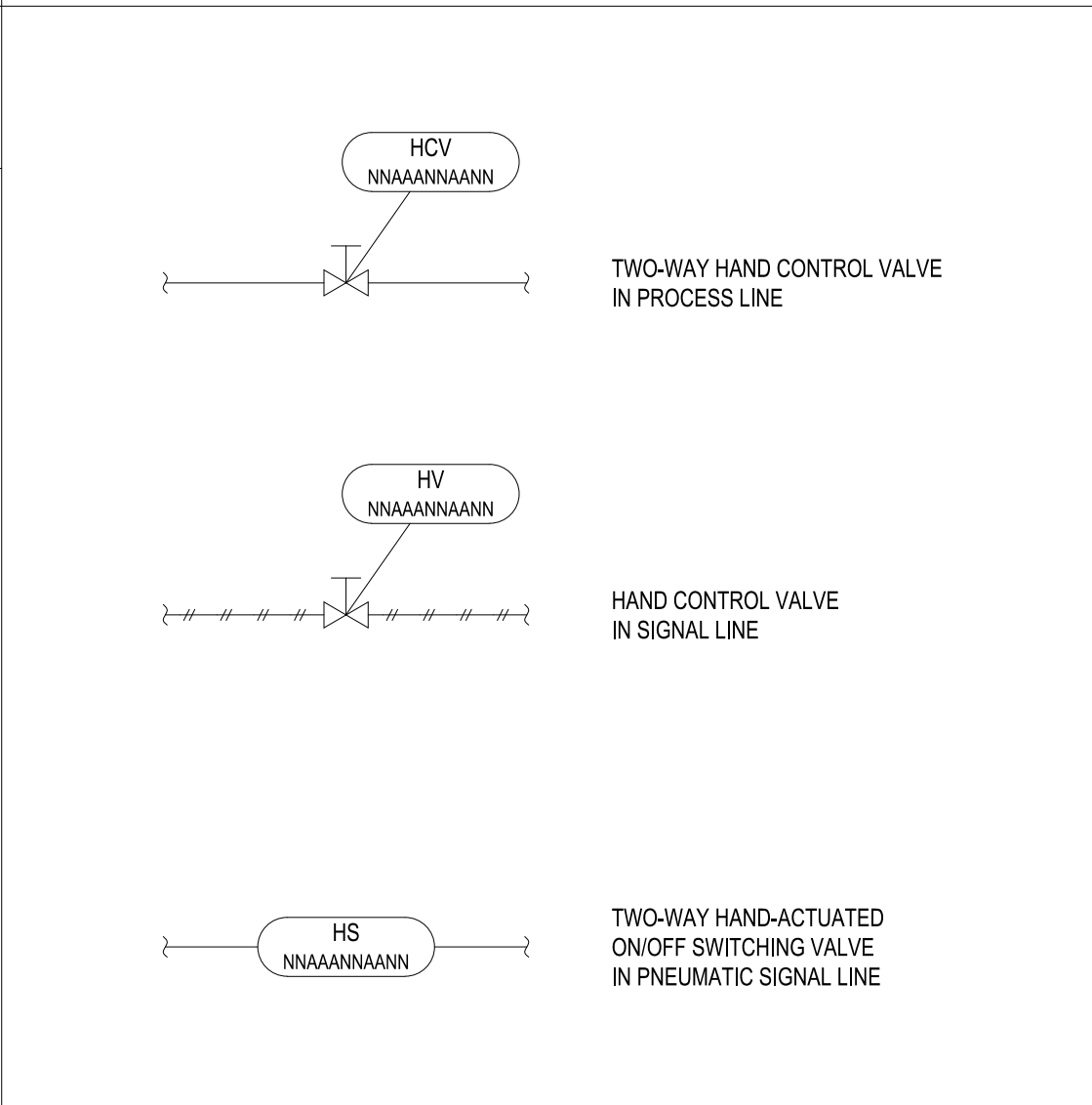
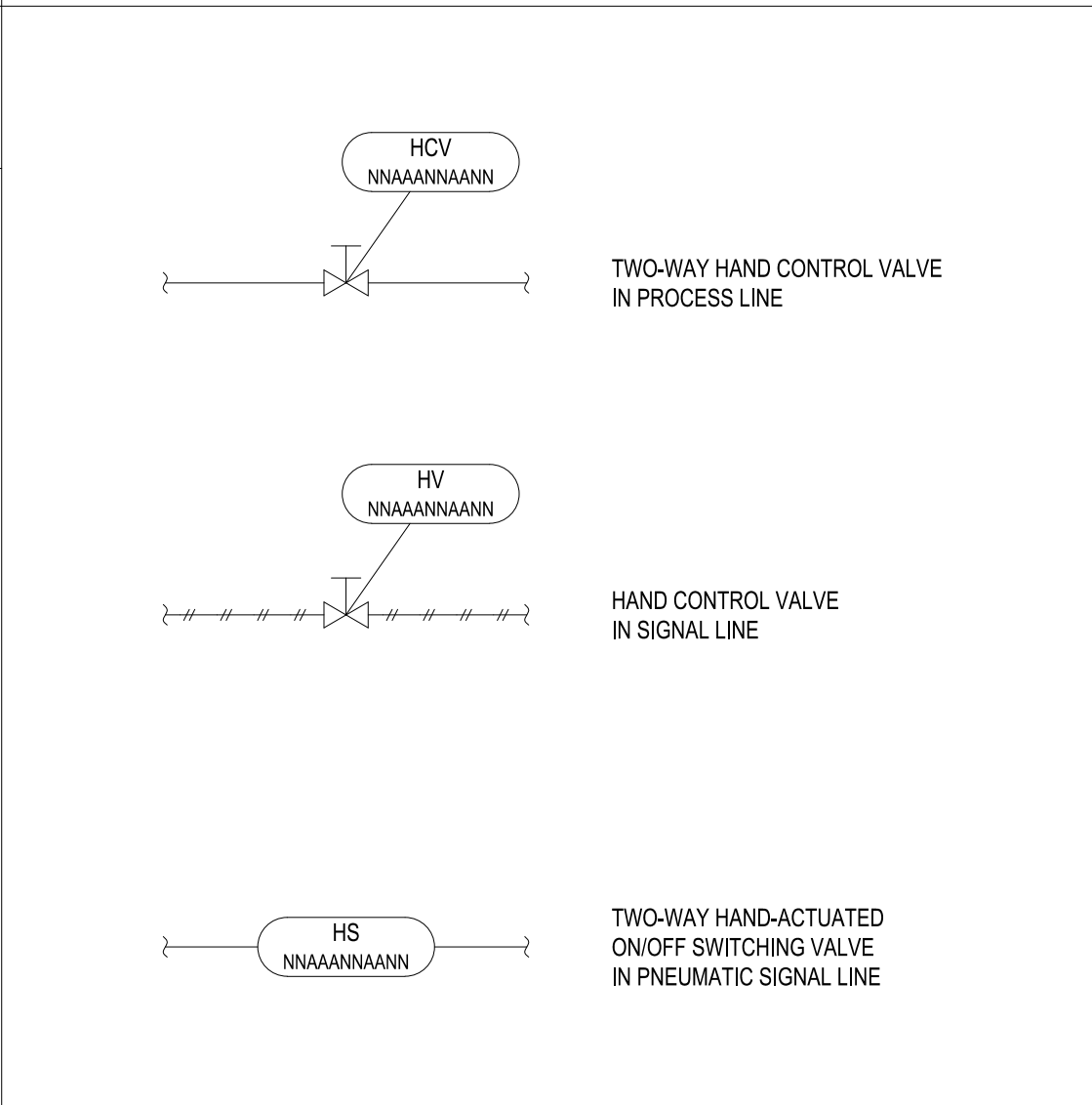
ESDV : EMERGENCY SHUT-DOWN VALVE  
SDV : SHUT-DOWN VALVE  
SET : SET VALUE  
ZSOC : POSITION SWITCH OPEN/CLOSE  
ZIOC : POSITION INDICATOR OPEN/CLOSE  
ZOCL : POSITION OPEN/CLOSE LAMP  
CC : CATIONIC CONDUCTIVITY  
SC : SPECIFIC CONDUCTIVITY  
XV : SHUT-OFF VALVE  
M : ELECTRIC MOTOR  
MH : MAN HOLE  
SMFCP : SETTLEMENT METERING FLOW COMPUTER PANEL  
BMFCP : BACK-UP METERING FLOW COMPUTER PANEL  
MECP: MAIN FIRE ALARM CONTROL PANEL  
ESEW : EMERGENCY SHOWER & EYE WASHER  
DCMS : DISTRIBUTED CONTROL AND MONITORING SYSTEM  
BMS : BURNER MANAGEMENT SYSTEM  
BPS : BOILER PROTECTION SYSTEM

[illegible]

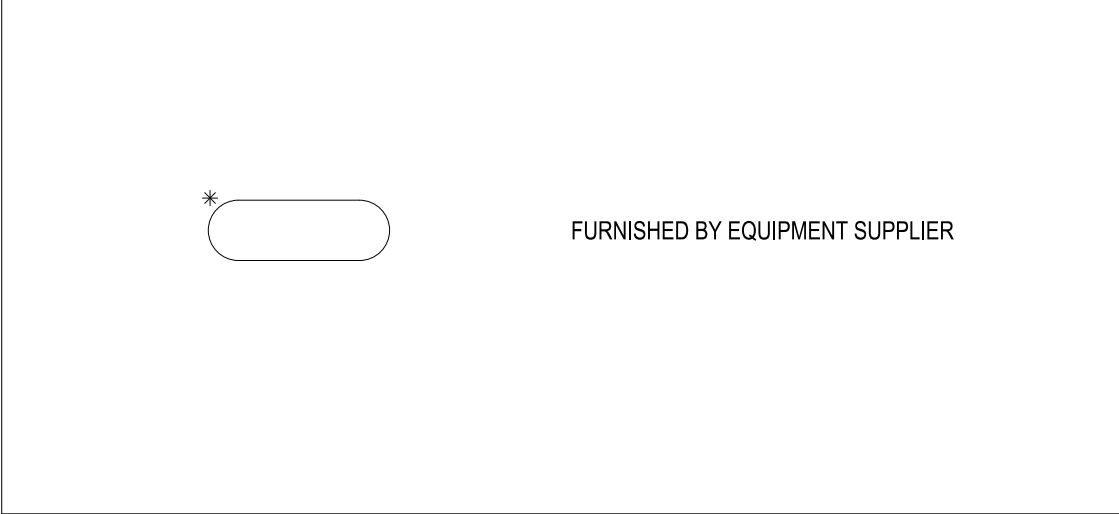
|                            |   |
|----------------------------|---|
| 3                          | 2 |
| SELF-ACTUATED DEVICES-TEMP |   |



## HAND ACTUATED DEVICES



|  |  |
|--|--|
|  |  |
|  |  |



## LOGIC & SEQUENCE CONTROL

|                                                                                       |                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>GENERAL</p> <ul style="list-style-type: none"> <li>- INTERCONNECTING LOGIC<br/>OR SEQUENCE CONTROL<br/>NORMALLY USED FOR LOCAL<br/>AUXILIARY CONTROL SYSTEM</li> </ul>                                                 |
|  | <p>NOT NORMALLY ACCESSIBLE TO OPERATOR</p> <ul style="list-style-type: none"> <li>- PACKAGED PROGRAMMABLE LOGIC<br/>CONTROLLER OR DIGITAL LOGIC CONTROLS<br/>INTEGRAL TO THE DISTRIBUTED CONTROL<br/>EQUIPMENT</li> </ul> |
|  | <p>NORMALLY ACCESSIBLE TO OPERATOR</p> <ul style="list-style-type: none"> <li>- PACKAGED PROGRAMMABLE LOGIC<br/>CONTROLLER OR DIGITAL LOGIC<br/>CONTROLS INTEGRAL TO THE DISTRIBUTED<br/>CONTROL EQUIPMENT</li> </ul>     |

| NOTES :                                                                                                                                              |             |                                                            |          |          |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|----------|----------|----------|
| <div style="border: 1px solid black; padding: 10px; display: inline-block;">FOR INTERNAL USE</div>                                                   |             |                                                            |          |          |          |
| <small>THIS DRAWING IS THE PROPERTY OF SAMSUNG C&amp;T CORPORATION<br/>IT IS NOT TO BE COPIED OR USED IN ANY WAY DETRIMENTAL TO THE COMPANY.</small> |             |                                                            |          |          |          |
|                                                                                                                                                      |             |                                                            |          |          |          |
|                                                                                                                                                      |             |                                                            |          |          |          |
|                                                                                                                                                      |             |                                                            |          |          |          |
|                                                                                                                                                      |             |                                                            |          |          |          |
|                                                                                                                                                      |             |                                                            |          |          |          |
|                                                                                                                                                      |             |                                                            |          |          |          |
| A                                                                                                                                                    | 19.SEP.2025 | FOR INTERNAL USE                                           |          |          |          |
| REV.                                                                                                                                                 | DATE        | DESCRIPTION                                                | PREPARED | REVIEWED | APPROVED |
| EMPLOYER                                                                                                                                             |             |                                                            |          |          |          |
| TENDERER                                                                                                                                             |             |                                                            |          |          |          |
|                                                                 |             | <b>SAMSUNG C&amp;T</b><br>Engineering & Construction Group |          |          |          |
| PROJECT NAME                                                                                                                                         |             |                                                            |          |          |          |
| AL NOUF1 PROJECT                                                                                                                                     |             |                                                            |          |          |          |
| DRAWING TITLE                                                                                                                                        |             |                                                            |          |          |          |
| P&ID FOR SYMBOLS & LEGENDS 3                                                                                                                         |             |                                                            |          |          |          |
| PROJECT DWG NO.                                                                                                                                      |             |                                                            | SHEET    | REV.     |          |
| D00P-00GEN00-M05-0004                                                                                                                                |             |                                                            | 1/1      | A        |          |
| PROJECT NO.                                                                                                                                          |             | REF. DRAWING NO.                                           | SCALE    | REF.CODE |          |
| D00P                                                                                                                                                 |             |                                                            | NONE     |          |          |

| NOTES :                                                                                                                                              |                  |                                                            |          |          |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------|----------|----------|----------|
|                                                                                                                                                      |                  |                                                            |          |          |          |
| <div style="border: 1px solid black; padding: 10px; display: inline-block;">FOR INTERNAL USE</div>                                                   |                  |                                                            |          |          |          |
| <small>THIS DRAWING IS THE PROPERTY OF SAMSUNG C&amp;T CORPORATION<br/>IT IS NOT TO BE COPIED OR USED IN ANY WAY DETRIMENTAL TO THE COMPANY.</small> |                  |                                                            |          |          |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
| A                                                                                                                                                    | 19.SEP.2025      | FOR INTERNAL USE                                           |          |          |          |
| REV.                                                                                                                                                 | DATE             | DESCRIPTION                                                | PREPARED | REVIEWED | APPROVED |
| EMPLOYER                                                                                                                                             |                  |                                                            |          |          |          |
| TENDERER                                                                                                                                             |                  |                                                            |          |          |          |
|                                                                 |                  | <b>SAMSUNG C&amp;T</b><br>Engineering & Construction Group |          |          |          |
| PROJECT NAME                                                                                                                                         |                  |                                                            |          |          |          |
| AL NOUF1 PROJECT                                                                                                                                     |                  |                                                            |          |          |          |
| DRAWING TITLE                                                                                                                                        |                  |                                                            |          |          |          |
| P&ID FOR SYMBOLS & LEGENDS 3                                                                                                                         |                  |                                                            |          |          |          |
| PROJECT DWG NO.                                                                                                                                      |                  |                                                            | SHEET    | REV.     |          |
| D00P-00GEN00-M05-0004                                                                                                                                |                  |                                                            | 1/1      | A        |          |
|                                                                                                                                                      |                  |                                                            |          |          |          |
| PROJECT NO.                                                                                                                                          | REF. DRAWING NO. |                                                            | SCALE    | REF.CODE |          |
| D00P                                                                                                                                                 |                  |                                                            | NONE     |          |          |

|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|---------------------------------------------------------------------------------------|-------------|------------------|------------------------------------------------------------|--|--|----------|----------|-----------|
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
| A                                                                                     | 19-SEP-2025 | FOR INTERNAL USE |                                                            |  |  |          |          |           |
| REV.                                                                                  | DATE        | DESCRIPTION      |                                                            |  |  | PREPARED | REVIEWED | APPROVED  |
| EMPLOYER                                                                              |             |                  |                                                            |  |  |          |          |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |
| TENDERER                                                                              |             |                  |                                                            |  |  |          |          |           |
|  |             |                  | <b>SAMSUNG C&amp;T</b><br>Engineering & Construction Group |  |  |          |          |           |
| PROJECT NAME                                                                          |             |                  |                                                            |  |  |          |          |           |
| AL NOUF1 PROJECT                                                                      |             |                  |                                                            |  |  |          |          |           |
| DRAWING TITLE                                                                         |             |                  |                                                            |  |  |          |          |           |
| P&ID FOR SYMBOLS & LEGENDS 3                                                          |             |                  |                                                            |  |  |          |          |           |
| PROJECT DWG NO.<br><br>D00P-00GEN00-M05-0004                                          |             |                  |                                                            |  |  |          | SHEET    | REV.      |
|                                                                                       |             |                  |                                                            |  |  |          | 1/1      | A         |
| PROJECT NO.                                                                           |             | REF. DRAWING NO. |                                                            |  |  |          | SCALE    | REF. CODE |
| D00P                                                                                  |             |                  |                                                            |  |  |          | NONE     |           |
|                                                                                       |             |                  |                                                            |  |  |          |          |           |

|                              |                  |       |           |
|------------------------------|------------------|-------|-----------|
| PROJECT NAME                 |                  |       |           |
| AL NOUF1 PROJECT             |                  |       |           |
| DRAWING TITLE                |                  |       |           |
| P&ID FOR SYMBOLS & LEGENDS 3 |                  |       |           |
| PROJECT DWG. NO.             |                  | SHEET | REV.      |
| D00P-00GEN00-M05-0004        |                  | 1/1   | A         |
| PROJECT NO.                  | REF. DRAWING NO. | SCALE | REF. CODE |
| D00P                         |                  | NONE  |           |

|                              |                  |       |           |
|------------------------------|------------------|-------|-----------|
| PROJECT NAME                 |                  |       |           |
| AL NOUF1 PROJECT             |                  |       |           |
| DRAWING TITLE                |                  |       |           |
| P&ID FOR SYMBOLS & LEGENDS 3 |                  |       |           |
| PROJECT DWG. NO.             |                  | SHEET | REV.      |
| D00P-00GEN00-M05-0004        |                  | 1/1   | A         |
| PROJECT NO.                  | REF. DRAWING NO. | SCALE | REF. CODE |
| D00P                         |                  | NONE  |           |

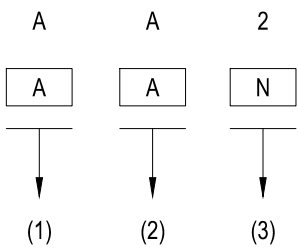
|                              |                  |       |           |
|------------------------------|------------------|-------|-----------|
| DRAWING TITLE                |                  |       |           |
| P&ID FOR SYMBOLS & LEGENDS 3 |                  |       |           |
| PROJECT DWG. NO.             |                  | SHEET | REV.      |
| D00P-00GEN00-M05-0004        |                  | 1/1   | A         |
| PROJECT NO.                  | REF. DRAWING NO. | SCALE | REF. CODE |
| D00P                         |                  | NONE  |           |

|                                              |                  |       |           |
|----------------------------------------------|------------------|-------|-----------|
| PROJECT DWG NO.<br><br>D00P-00GEN00-M05-0004 |                  | SHEET | REV.      |
|                                              |                  | 1/1   | A         |
| PROJECT NO.                                  | REF. DRAWING NO. | SCALE | REF. CODE |
| D00P                                         |                  | NONE  |           |



TYPICAL IDENTIFICATION SYSTEM FOR  
PIPE SIZE,  
CLASS, MATERIAL AND INSULATION

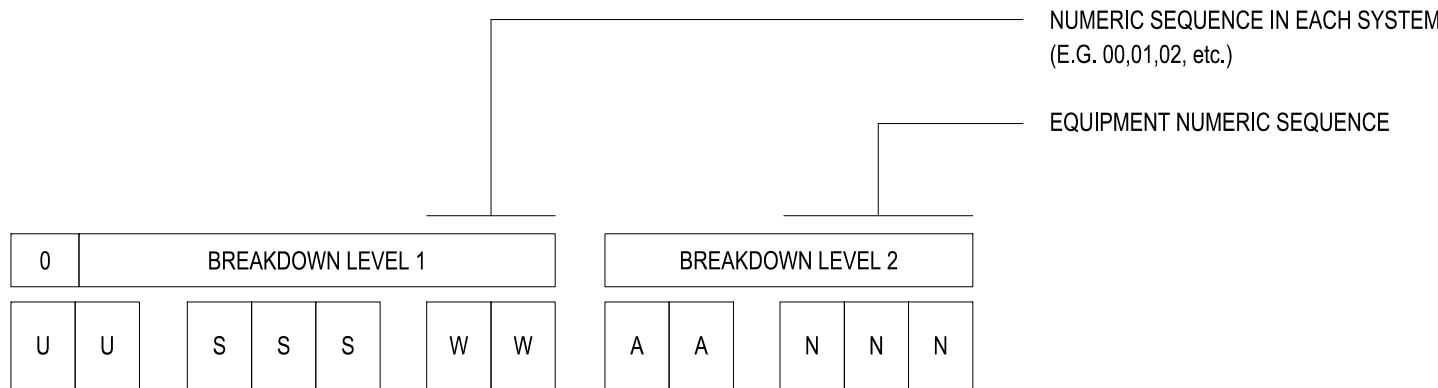
EXAMPLE OF  
APPLICATION :



- (1) CLASSRATING  
A : 4500# or 2500#SP (ASME B16.5)  
B : 2500# (ASME B16.5)  
C : 1500#SP (ASME B16.5)  
D : 1500# (ASME B16.5)  
E : 900# (ASME B16.5)  
F : 600# (ASME B16.5)  
G : 300# (ASME B16.5)  
H : 150# (ASME B16.5)  
J : 125# (ASME B16.1)  
X : Special as Designated  
(2) PIPING MATERIAL  
A : ALLOY STEEL (Cr-Mo)  
B : CARBON STEEL (CS)  
C : STAINLESS STEEL (304 or 304L)  
D : STAINLESS STEEL (316 or 316L), DUPLEX STAINLESS STEEL  
E : CAST IRON or DUCTILE IRON (CI or DI)  
F : GALVANIZED CARBON STEEL  
G : COPPER  
H : PVC or CPVC  
J : PVDF  
K : PLASTIC LINED CARBON STEEL  
L : HIGH DENSITY POLYETHYLENE (HDPE)  
M : POLYPROPYLENE (PP)  
R : FRP, GRP, GRE, GRV, etc.  
S : CONCRETE  
(3) SERIAL NO.

TYPICAL IDENTIFICATION SYSTEM FOR  
PIPE LINE, VALVE, EQUIPMENT AND FILTER, SCREEN,STRAINER, etc.

TAG NUMBERING SYSTEM IS BASED ON THE KKS NUMBERING SYSTEM. (DOC. NO.: 19K9-00GEN00-G02-0002)

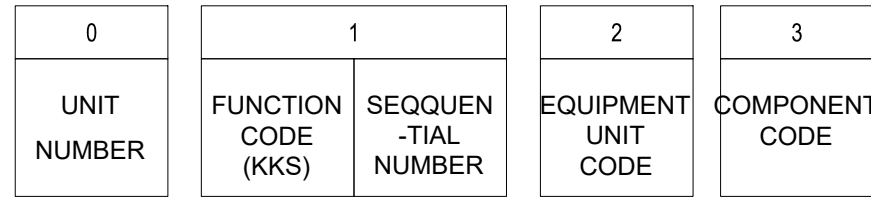


UNIT IDENTIFICATION NUMBERS

|             | PLANT UNIT                                   | "UU" CHARACTER<br>DESIGNATION |
|-------------|----------------------------------------------|-------------------------------|
| POWER PLANT | POWER PLANT COMMON                           | 0                             |
|             | FIRST GROUP COMMON (INCL. FIRST GROUP STG)   | 10                            |
|             | FIRST GROUP #1 GTG / HRSG                    | 11                            |
|             | FIRST GROUP #2 GTG / HRSG                    | 12                            |
|             | SECOND GROUP COMMON (INCL. SECOND GROUP STG) | 20                            |
|             | SECOND GROUP #1 GTG / HRSG                   | 21                            |
|             | SECOND GROUP #2 GTG / HRSG                   | 22                            |
|             |                                              |                               |

THE BASIS FOR THE DESIGNATION OF INSTALLATIONS AND SUBSYSTEMS  
IN POWER PLANTS.

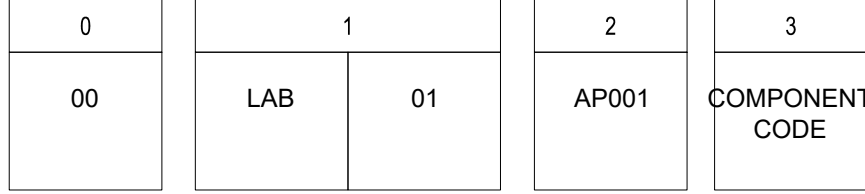
SERIAL NO. OF  
BREAKDOWN LEVEL



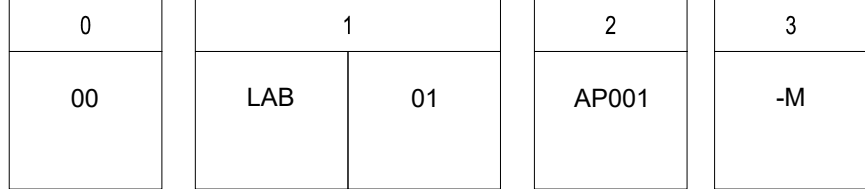
DESIGNATION OF  
BREAKDOWN LEVEL

EXAMPLE OF  
APPLICATION:

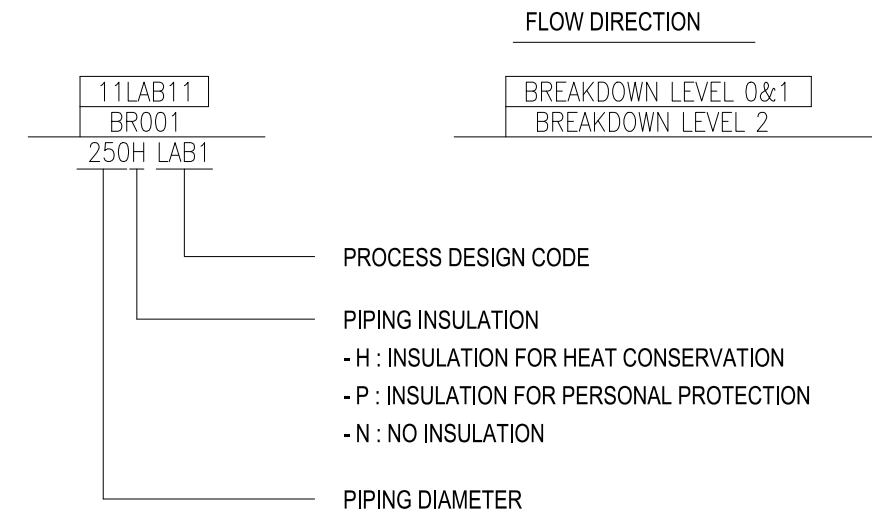
20LAB31AP001 (COGENERATION FEEDWATER PUMP A)



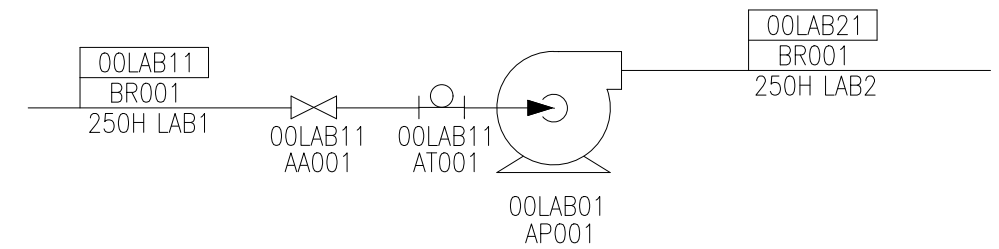
20LAB11AP001-M (COGENERATION FEEDWATER PUMP A MOTOR)



PIPING DESIGNATION FLAG  
(POINTING IN FLOW DIRECTION INCLUDING PIPING DIAMETER,  
PROCESS DESIGN CODE, PIPING INSULATION)



EXAMPLE FOR LINE VALVE, EQUIPMENT AND STRAINER IDENTIFICATION



GENERAL NOTES :

1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE  
ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL  
DESIGN PHASE.

NOTES :

FOR INTERNAL USE

THIS DRAWING IS THE PROPERTY OF SAMSUNG C&T CORPORATION.  
IT IS NOT TO BE COPIED OR USED IN ANY WAY DETRIMENTAL TO THE COMPANY.

|           |             |                  |          |          |          |
|-----------|-------------|------------------|----------|----------|----------|
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
|           |             |                  |          |          |          |
| A         | 19.SEP.2025 | FOR INTERNAL USE |          |          |          |
| REV.      | DATE        | DESCRIPTION      | PREPARED | REVIEWED | APPROVED |
| EMPLOYER: |             |                  |          |          |          |

TENDERER  
**SAMSUNG** **SAMSUNG C&T**  
Engineering & Construction Group

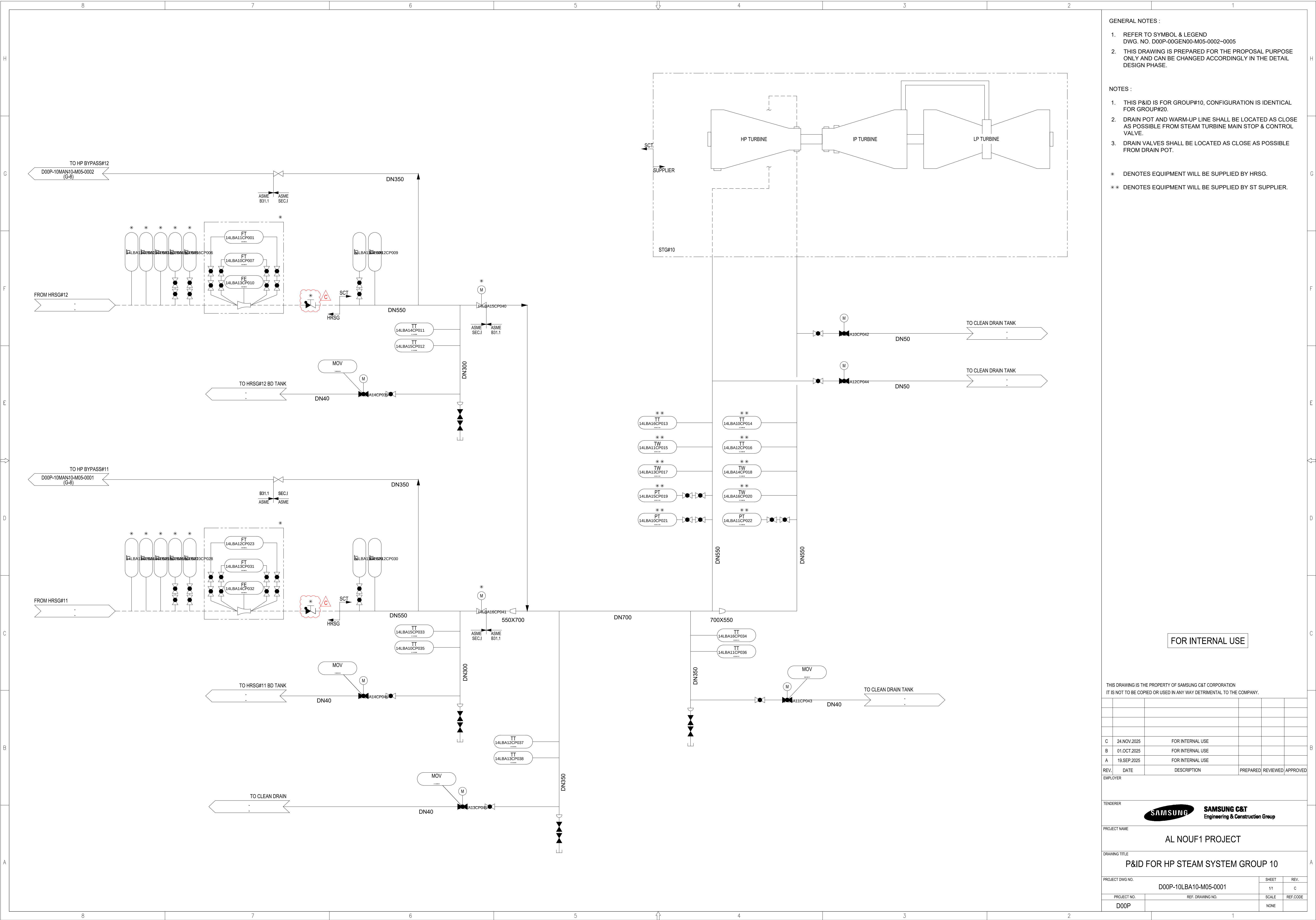
PROJECT NAME  
**AL NOUF1 PROJECT**

DRAWING TITLE  
**P&ID FOR SYMBOLS & LEGENDS 4**

|                       |                  |       |           |
|-----------------------|------------------|-------|-----------|
| PROJECT DWG NO.       |                  | SHEET | REV.      |
| D00P-00GEN00-M05-0005 |                  | 1/1   | A         |
| PROJECT NO.           | REF. DRAWING NO. | SCALE | REF. CODE |
| D00P                  |                  | NONE  |           |

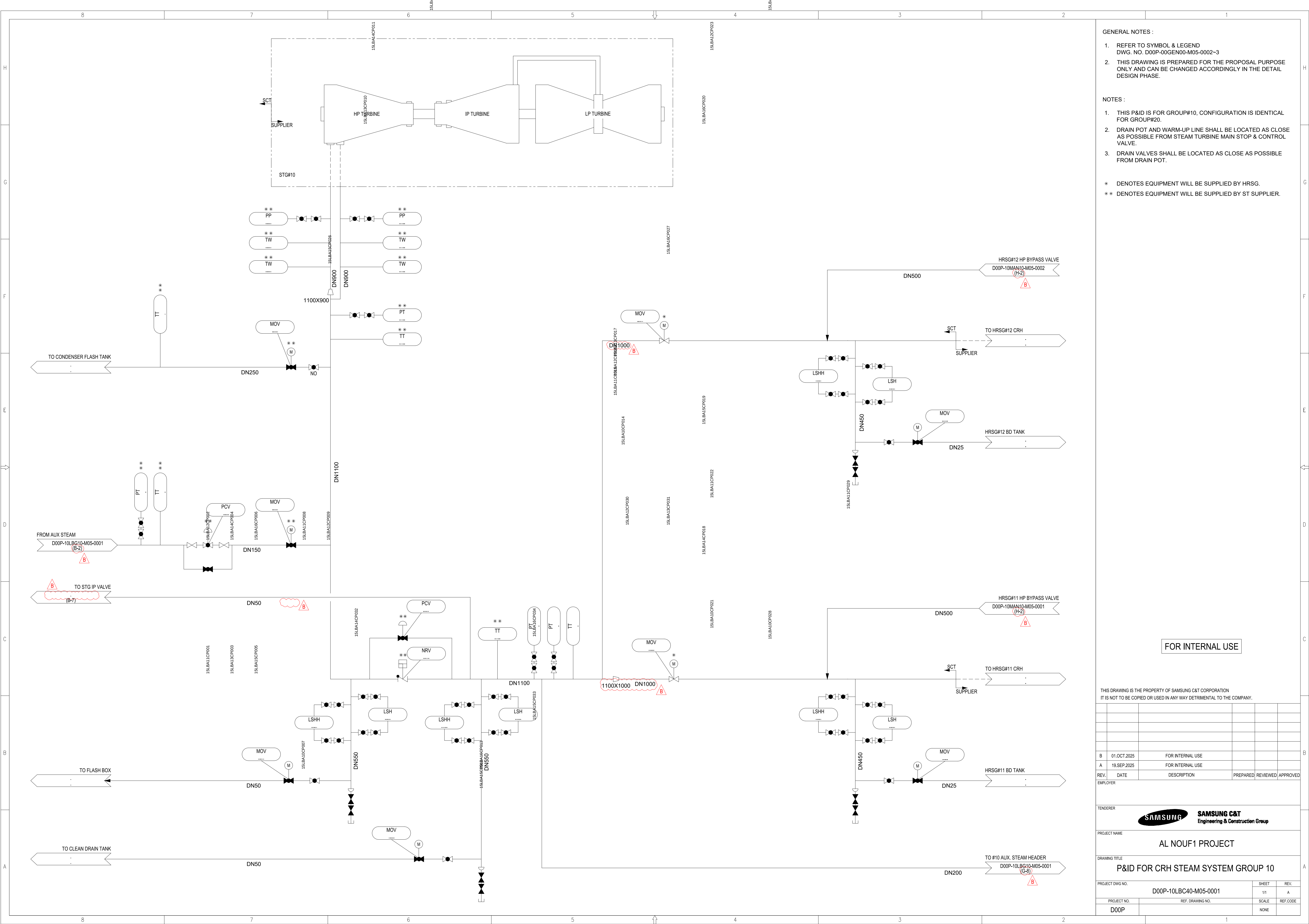
| NUMBER<br>RANGE | VALVES                                                  | PIPE BR                                                | INSTRUMENTATION<br>CIRCUITS                                                                      |
|-----------------|---------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 001-100         | -VALVES IN MAIN PIPES<br>-VALVE GENERAL                 | -MAIN PIPES                                            | -TRANSMITTER                                                                                     |
| 101-190         | -CONTROL VALVES / DAMPERS                               |                                                        | -SWITCH                                                                                          |
| 191-199         | - SAFETY VALVES                                         | -SUCTION AND PRESSURE RELIEF<br>LINES ON SAFETY VALVES |                                                                                                  |
| 201-250         | -TCE IS USING FOR MOV IN MAIN<br>LINES                  | -NOT USED                                              | -FOR TESTING EQUIPMENTS                                                                          |
| 251-299         | -SHUT OFF VALVE MAIN LINE                               |                                                        |                                                                                                  |
| 301-399         | -INSTRUMENT ISOLATION                                   | -PRESSURE LINES FOR<br>INSTRUMENTS                     | -NOT USED                                                                                        |
| 401-499         | -VALVES IN DRAINS AND<br>FLUSHING PIPES                 | -DRAINS, FLUSHING PIPES,<br>PRESSURE SUPPERSSION PIPES | -ACCEPTANCE TESTS<br>-TEST THERMOWELL                                                            |
| 501-599         | -VALVES FOR VENTING                                     | -VENTING                                               | -LOCAL MEASURING CIRCUITS<br>-GAUGES                                                             |
| 601-699         | -VALVES FOR SAMPLING AND<br>DOSING                      | -SAMPLING AND DOSING PIPES                             | -FLOW ELEMENT                                                                                    |
| 701-799         | -VALVES FOR INTERNAL<br>CONTROLS                        | -FOR INTERNAL CONTROLS                                 | -GROUPED                                                                                         |
| 801-899         | 1. TCE IS USING FOR MOV IN DRAIN LINES<br>2. STEAM TRAP | -NOT USED                                              | -GROUPED                                                                                         |
| 901-999         | -TCE IS USING FOR MOV IN VENT<br>LINES                  | -NOT USED                                              | -GROUP MEASURING CIRCUITS<br>(E.G. MULTI POINT SELECTOR<br>SWITCHES, MULTI CHANNEL<br>RECORDERS) |





- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND  
DWG. NO. D00P-00GEN00-M05-0002~0005
  - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
  - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
  - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- \* DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
- \*\* DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.





- GENERAL NOTES :
1. REFER TO SYMBOL & LEGEND  
DWG. NO. D00P-00GEN00-M05-0002~3
  2. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
1. THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
  2. DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
  3. DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- \* DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.  
\*\* DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

FOR INTERNAL USE

THIS DRAWING IS THE PROPERTY OF SAMSUNG C&T CORPORATION  
IT IS NOT TO BE COPIED OR USED IN ANY WAY DETRIMENTAL TO THE COMPANY.

| REV.     | DATE        | DESCRIPTION      | PREPARED | REVIEWED | APPROVED |
|----------|-------------|------------------|----------|----------|----------|
| B        | 01.OCT.2025 | FOR INTERNAL USE |          |          |          |
| A        | 19.SEP.2025 | FOR INTERNAL USE |          |          |          |
| EMPLOYER |             |                  |          |          |          |

TENDERER  
**SAMSUNG** **SAMSUNG C&T**  
Engineering & Construction Group

PROJECT NAME  
AL NOUF1 PROJECT

DRAWING TITLE  
P&ID FOR CRH STEAM SYSTEM GROUP 10

| PROJECT DWG. NO.      | PROJECT NO. | REF. DRAWING NO. | SHEET | REV. |
|-----------------------|-------------|------------------|-------|------|
| D00P-10LBC40-M05-0001 |             |                  | 1/1   | A    |
| D00P                  |             |                  | NONE  |      |